

Characterize Before You Loop: What Transfers From a Zero-Shot Proposer’s Template and What Does Not

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Abstract

A weak-tier language model proposer asked in a zero-shot call to pack circles in the unit square concentrates on a grid-plus-filler template, and a companion submission gives the closed form naming its modal output. This paper measures what survives a change of setting. Eight registered arms change four things: the parent, the iteration, the serving path with its reasoning setting, and the vendor. One conditioning step dissolves the anchor. Under a better in-family parent, 26 of 26 valid outputs return that parent’s score, 25 within 10^{-6} and one 1.2×10^{-6} above it, which entails an anchor-rate of 0 of 26; under its own anchor as parent, 7 of 28 stay on it, against a registered floor of 50%, and 17 move above it; from an off-family parent the template returns in 20 of 34. Five loop generations take the on-prediction rate from 11 of 17 to 11 of 49, clearing the family argmax nowhere. The same prompts on a pinned serving path return 0 of 105 valid packings, and extended thinking restores validity, 12 of 14 against 0 of 13. At a second vendor the pooled bar is met, 46 of 80, and the discriminating cells miss it, 11 of 43. The anchor does not transfer; the family argmax does, as a ceiling no arm clears at 10^{-9} . At the primary 10^{-6} tolerance, 18 conditioned outputs exceed it by at most 1.5×10^{-6} , 17 of them returning a parent whose seven-decimal coordinates already exceed it. Every registered arm is reported, whichever way each went.

1 Introduction

Ask a language model to place 21 circles in a unit square to maximize the sum of their radii, and it will not search. It reaches for the nearest square grid, five by five ($k = 5$) at radius $1/(2k)$, and truncates it to 21 circles, leaving four cells empty. A companion submission (Anonymous authors, 2026) gives that behaviour a closed form, one that names the modal output in advance. It reports that form as the empirical mode at all seven N tested and four of the five held out (both sets in Table 2), and under a perturbed container and paraphrase. This paper’s own generation-0 rows, sampled fresh under arm L’s registration with the companion’s prompt bytes on the same instrument (Anonymous authors, 2026), put the predicted value on 4 of 9 valid outputs at $N = 13$ and on 7 of 8 at $N = 31$.

A characterization that sharp invites an inference: that the value a proposer emits in a zero-shot call is what it will contribute inside a discovery loop, so the loop can be reasoned about without being run. LLM-driven discovery systems in the FunSearch line (Romera-Paredes et al., 2024; Novikov et al., 2025) do not call their proposer zero-shot. They condition it on a parent and iterate it against an archive. Each of those is a change of setting, and so is the serving path (Anonymous authors, 2026); each is a place the characterization can fail.

This paper measures the failures. Eight registered arms change four things about the setting between them: arm MU the parent the proposer is conditioned on, arm L the iteration, feeding a scored proposal back as the next parent, arms P and P-D the serving path and the reasoning setting on it, and arms GM, GM2, GM3 and V the vendor, GM2 and GM3 differing only by output budget at that vendor, 4,096 tokens against 16,384. Each registered prediction, verdict and falsifier appears below with its outcome, met or not.

Claim. The template a weak-tier LLM proposer concentrates on in zero-shot circle packing is contingent on the serving path and the request parameters it carries, the reasoning setting included. Arm P-D isolates that setting at one cell on 14 and 13 rows. A zero-shot characterization of a proposer therefore licenses

no prediction about the same proposer inside a discovery loop without a measurement at that loop’s own conditions. What fails to transfer is the anchor; what transfers is the family argmax, as a ceiling (terms in Table 2). No valid output clears it in any arm, 0 of 49 in arm L and 0 of 63 in arm MU at 10^{-9} , and 27 of 43 valid outputs at the second vendor’s discriminating cells sit on it. At the primary 10^{-6} tolerance, 18 arm MU outputs exceed the argmax by at most 1.5×10^{-6} , each on slightly overlapping circles (§4). Table 1 gives each arm that returned scoreable data, its registered bar and its outcome. Two arms carry no row there: arm GM finished UNDERPOWERED at 20 valid of 140 under both of its serving-path accountings, and arm GM2 returned 0 of 140 parseable at a 4,096-token output budget, the failure arm GM3 then diagnosed at 16,384.

Table 1: Six of the eight registered arms: what each changed about the setting, the bar registered before sampling, the measured outcome and the verdict. Bracketed ranges are 95% Wilson intervals on the rate they follow. L1 to L4 are the loop arm’s four registered predictions, named in §5; all prediction and falsifier codes are expanded in each arm’s own section and listed in full in Supplement S1.

Arm	What changed	Registered bar	Outcome	Verdict
MU	The parent the proposer is conditioned on	Under a better in-family parent, anchor-rate at most 20% and keep-or-improve at least 50%	26 of 26 keep or improve [87–100%], 25 within 10^{-6} of the parent’s score and one 1.2×10^{-6} above it, so the anchor-rate there is 0 of 26 [0–13%]; under the model’s own anchor, 7 of 28 keep it [13–43%] and 17 move above it	The rule reduces to one condition under the better parent; the own-anchor rate misses P-MU1’s 50% bar, so the template is not self-stable; F-MU1 is that rule’s negation and adds no second result
L	Five archive-and-mutate generations	Pooled conditioned on-prediction rate below the generation-0 rate in both cells (L2); falsifier F-L1	$N = 13$: 4 of 9 at generation 0, 5 of 20 conditioned. $N = 31$: 7 of 8, then 6 of 29. Pooled 11 of 17 [41–83%], then 11 of 49 [13–36%]	L2 holds; L1 and L3 not met; L4 open (0 of 49 clear the family, [0–7%])
P	The serving path: the bare pinned chat-completions endpoint	A floor of five valid outputs per cell	0 of 105 valid at the square cells [0–4%]; 4 of 75 at the held-out cells [2–13%]	Every direct-emission prediction UNSCOREABLE: no cell reaches the floor
P-D	Extended thinking on against off, same pinned path	P-PD1 (the reasoning budget is the cause) against P-PD2 (the context is) against P-PD3 (anything else)	Thinking on, 12 of 14 valid with the template the plurality value at 5 of 12 [19–68%]; off, 0 of 13 [0–23%]	P-PD1 holds; extended thinking is the component
GM3	A second vendor’s gemma at a 16,384-token budget	Pooled rate at least 30%, and the predicted value modal in 5 of 5 scoreable cells	46 of 80 pooled (57.5%, [47–68%]); modal in 2 of 5; 11 of 43 at the discriminating cells (26%, [15–40%])	The pooled bar is met; the discriminating cells, the only ones that test anchoring, miss it
V	Thirteen free-tier aliases	At least 10 of a model’s 25 invocations valid, then V1 at least 50% and V2 at most 10%	Ten aliases below the floor; two aliases meet the V1 pooled bar at point estimate, on 8 of 12 [39–86%] and 11 of 18 [39–80%] of their valid outputs, and a third does not, at 0 of 15 [0–20%]	The boundary is located, not attributed

Contributions.

1. *A measurement of transfer, not an argument about it.* Each change of setting was given its own registered arm, with numeric thresholds and a falsifier fixed before the first invocation. All eight are reported and every deviation is tabled.

2. *A named instrument component, at one cell.* On the pinned path at $N = 13$, the collapse is not caused by the absence of a system prompt: adding a system line leaves it in place, at 0 of 13 valid. One registered condition (P-D) and two post hoc diagnostics separate the candidates, and extended thinking is the one that carries the template there.
3. *A cross-vendor boundary that is located but not attributed.* Two second-vendor measurements disagree, with model variant and instrument confounded.

Scope. The proposer under test is one vendor’s weak-tier alias (**claude-haiku-4.5**), except where another vendor is named. The conditioning arm is run at three trap cells, $N = 13, 21$ and 31 , and the loop arm at two of them, $N = 13$ and 31 .

What is not claimed. Nothing about the companion paper’s numerical baseline, about which constructions exceed the best value the recipe family can express, or about the vendor’s stronger mid-tier alias; those are its results, cited and not restated (Anonymous authors, 2026). Nothing about mechanism inside the weights, about what any named discovery system does at generation zero, or about whether its diversity machinery (Mouret & Clune, 2015; Lehman & Stanley, 2011) repairs or explores. Preregistration is an adopted standard, not a contribution (§8).

2 The template and its closed form

This section restates the object under test. Every definition in it is the companion paper’s (Anonymous authors, 2026); nothing here is a result of this paper.

The benchmark. Maximize the sum of radii of N non-overlapping circles in the unit square. Feasibility is decidable exactly from emitted coordinates, so scoring needs no human. The direct-emission prompt fixes the count, states containment and non-overlap, forbids code, and demands a raw Python list of `[x, y, r]` triples; the program channels replace the code-free clause with a program contract. The prompt and its digest are in the supplement.

The recipe family. The **recipe family** is the parametric set of grid-plus-filler constructions; a **template** is one member, and the **anchor** is the template the distribution concentrates on at a given N .

A $k \times k$ grid places one circle per cell center with $r_{\text{grid}} = 1/(2k)$. Then m fillers sit on interior grid vertices, tangent to the four grid circles around them, with $r_{\text{filler}} = (\sqrt{2} - 1)/(2k)$ and $0 \leq m \leq (k - 1)^2$. Hence $V(k, m) = k/2 + m(\sqrt{2} - 1)/(2k)$. When $N < k^2$ the model does not fall back to the next smaller complete grid, the $(k - 1) \times (k - 1)$ lattice at radius $1/(2(k - 1))$ with fillers on its interior vertices. It *truncates*, laying out the $k \times k$ lattice, occupying N cells and keeping the radius, which gives $T(k, N) = N/(2k)$.

The **family argmax** at N is the largest value the family can express there, recomputed in closed form over every admissible (k, m) . Inside a trap zone it is the larger of $V(k^* - 1, N - (k^* - 1)^2)$, the drop-and-fill construction, and the truncated anchor $T(k^*, N) = N/(2k^*)$. At $N = 35$ the truncated grid is the larger of the two, which is why 35 is not a discriminating cell.

Independent re-score. Every count in this paper was recomputed by a second implementation the authors wrote for this submission (`arm_transfer_independent_rescore.py` in the artifact), with its own parser, its own validity scorer and a linear program blind to the recipe, importing nothing from the arms’ scoring code. It re-derived the closed-form value and the family argmax at all twelve N , agreeing to at most 3.6×10^{-15} . Of 135 cell-level counts, 129 agree; the six that differ are three arm L clearance counts at $N = 31$, two arm GM validity counts and arm P-D’s sampled-row count, each with its cause in Supplement S1. The re-score also found the arm MU bucket error recorded as corrections-ledger item 36, and §4 carries the corrected figures.

The order rule and the branch rule. Write $k^*(N) = \text{round}(\sqrt{N})$ for the nearest-square order. This is a definition, not a hypothesis. The falsifiable content is the **branch rule**: $k^{*2} \leq N$ gives extend with $m = N - k^{*2}$ fillers (converge), and $k^{*2} > N$ gives truncate the k^* -grid (trap). Substituting k^* into the branch condition gives the *trap zones* $N \in [k^2 - k + 1, k^2 - 1]$, which over the sweep range are $[13, 15]$, $[21, 24]$, $[31, 35]$, $[43, 48]$ and $[57, 63]$. At a converge cell the extend prediction is the argmax, so every discriminating cell is a trap cell, and the surviving in-family prediction is the truncate branch $T(k^*, N) = N/(2k^*)$.

Notation and conventions. Table 2 restates the companion paper’s cell sets and terms.

Table 2: Terms used throughout this paper, each fixed here and used in this sense only.

Term	Meaning
on-prediction family argmax	emitted sum within 2×10^{-3} of the registered predicted value the highest-valued member of the recipe family at that N : inside a trap zone, the drop-and-fill $V(k^*-1, m)$ wherever it beats the truncated anchor. This paper gives it no other name
square, held-out and code cell	the three cell sets: a <i>square cell</i> is an N addressed with the bare direct-emission prompt, $N = 13, 17, 21, 31, 35, 37, 43$; a <i>held-out cell</i> is an N the companion paper kept out of its fit and this paper reuses, $N = 50, 58, 62, 65, 75$; a <i>code cell</i> is an N addressed under the program contract instead, $N = 13, 21, 31$
discriminating cell	N where prediction $\neq$ family argmax; only these can distinguish anchoring from family search
validity trap cell	non-overlap and containment at 10^{-6} (primary) and 10^{-9} (logged), both always reported an N inside a trap zone, where the branch rule truncates; the discriminating cells are the trap cells whose truncated anchor sits below the family argmax
tier	nominal capability class of the serving alias addressed: here the weak tier is the vendor’s Haiku alias (claude-haiku-4.5)
instrument	the proposer’s serving path with the request parameters it carries, the reasoning setting included. Two are used here, an agent runtime and a pinned API, described in §3; they are never pooled, and the word is used in this sense only
effort	the size of the reasoning budget a request asks the vendor for. Where a request names none the vendor’s routing layer supplies its own default, which is the setting arm P-D switches on
quota	the vendor-side cap on requests per interval; a key that reaches it returns no completions until the interval resets
UNSCOREABLE evaluability floor	cell with fewer valid samples than its arm’s registered floor; claims nothing the minimum valid-sample count a cell needs before its verdict counts, fixed at each arm’s own registration and <i>not</i> common across arms; the per-arm values are in §3
template anchor $T(k^*, N)$	the truncated-grid value at the nearest-square order, $N/(2k^*)$; at a discriminating cell it sits below the family argmax
k -inheritance	an output whose dominant circle radius gives the same grid order k as its parent’s; counted per parent condition in the arm MU table
located but not attributed	a difference measured between conditions differing in more than one way, so that the difference is established and its cause is not

3 Registration, instruments, scoring, floors

Registration protocol. For each cell the prompt is pinned and its SHA-256 digest written to disk first. The $N = 13$ square prompt has a SHA-256 digest beginning **32db485b**, given in full in the supplement. Predictions live in the header of each arm’s analysis script or in a standalone registration file. Competing predictions, numeric thresholds, tie conventions and a falsifier are fixed before the first invocation. Raw outputs are stored verbatim, failures included. Scoring is deterministic and local. Each arm writes its rows to a JSONL ledger. The supplement carries every arm’s ledger, every departure from a registration, and the per-cell tables for the serving-path and second-vendor arms. Every reported verdict was registered before its arm’s first scored run; post hoc readings are labelled where they appear and listed together in §9.

Two instruments, never pooled. The first is an agent runtime. It does not expose decoding parameters, so every runtime row is a distributional claim over an observed sampling regime. It dispatches each call to a *subagent*, the runtime’s worker process that receives the prompt, and that process inherits instruction files outside the task prompt, which the digests do not lock and a replicator cannot reconstruct; it also publishes no weight hash, so the alias-to-weights binding is a vendor promise. The second is a pinned API, the vendor’s chat-completions endpoint reached through OpenRouter, at temperature 1.0 with no system prompt. It exposes and pins temperature and carries no inherited context. Arms MU and L run on the runtime. Arms P and P-D run on the pinned path. Arms GM, GM2 and GM3 run on a second vendor’s first-party API, and arm V runs on free-tier aliases across seven attributed vendors. No count in this paper pools across instruments.

Scoring conventions, fixed in advance. Validity is reported at 10^{-9} and 10^{-6} , both logged, with 10^{-6} primary. Proposers print six to eight decimals, so an eight-decimal tangency misses contact by about 5×10^{-9} , while 10^{-6} sits far below the gap of about 10^{-2} between the competing constructions a cell can express. Value matching uses a 2×10^{-3} window. That window never carries a clearance verdict. A proposal clears the family argmax when its verified sum strictly exceeds the closed-form argmax at the tolerance named with the count. Completions are parsed with `ast.literal_eval` after stripping code fences; one that fails to yield a list of N strictly positive numeric triples is a logged parse failure, scored invalid. Every on-prediction rate is computed over valid outputs and so conditions on execution success. Per-cell validity denominators are reported beside every rate, except in arm MU. That arm registered its anchor-rate and its keep-or-improve rate pooled across its three cells, and registered P-MU4 alone per cell, so its table gives pooled counts for the first two and per-cell counts for P-MU4.

Evaluability floors, per arm. The evaluability floor is the minimum valid-sample count a cell needs before its verdict counts, fixed at each arm’s own registration. The second-vendor chain uses 3: a cell below it is UNSCOREABLE for mode analysis and claims nothing in either direction. Arms P and P-D use 5. Arm V uses a per-model floor of at least 10 of a model’s 25 registered invocations valid at 10^{-6} , below which the model is labelled BELOW-FLOOR and carries no anchoring claim in either direction. Arm L’s registered floor is a two-part lineage rule: at least 3 valid proposals at generation 0, and at least 1 in at least 3 of the 5 conditioned generations. All four lineages clear it, 13-GREEDY on 5 valid at generation 0 and on valid outputs in 3 of its 5 conditioned generations, so its 4 valid conditioned proposals are not the quantity the floor tests.

The eight arms. All eight were run once. Arms P and P-D are reported in full in both submissions; the other six are reported here and appear in the companion only as a pointer. Each arm’s design is stated where the arm is reported; Supplement S1 tables all eight.

4 One conditioning step: arm MU

Arm MU is the smallest version of the change a discovery loop makes when it hands its proposer a parent: one parent, one instruction, no iteration.

Design. Arm MU issues one-parent mutation calls. The call is the bare prompt plus an existing packing plus the instruction to modify it so that the sum of radii increases. Three parents are used per cell. A_anchor is a packing built at the model’s own unconditioned anchor value $T(k^*, N)$. B_rival is a packing built at the family argmax $V(k^*-1, m)$, which at these cells is provably better than the anchor. $C_offfamily$ is a diagonal-chain packing scoring far below either. The cells are $N = 13, 21$ and 31 , at $n = 15$ per condition-cell, which is 135 invocations. The arm was registered before sampling with numeric decision thresholds, a power analysis, tie-against conventions, and a falsifier whose written consequence, if it fired, was to scope the anchoring result to unconditioned calls only.

Registered predictions. P-MU1 predicted an anchor-rate under A_anchor of at least 50%. P-MU2 predicted an anchor-rate under B_rival of at least 35%. P-MU3 predicted k -inheritance under B_rival below 50%. P-MU4 predicted that under $C_offfamily$ the modal valid output sits on the k^* -grid, scored per cell. The registered decision rule reads DISSOLVES when the pooled anchor-rate under B_rival is at most 20% and the keep-or-improve rate is at least 50%. Falsifier F-MU1 fires on the same condition.

The rule’s two conditions under B_rival . The pooled anchor-rate under B_rival is 0 of 26 valid outputs (95% Wilson 0–13%), below the registered ceiling of 20% and P-MU2’s 35% floor, and keep-or-improve is 26 of 26 (87–100%), above its floor of 50%. The two are not independent here. The B_rival parent outscores the anchor $T(k^*, N)$ by 0.151 at $N = 13$, 0.159 at $N = 21$ and 0.165 at $N = 31$, each more than seventy times the 2×10^{-3} matching window, so an output that keeps or improves on that parent cannot sit on the anchor: 26 of 26 keep-or-improve entails the 0 of 26, and the rule reduces to one condition. Keep-or-improve is almost entirely keep. Of the 26, 25 return the parent’s score within 10^{-6} and one exceeds it by 1.2×10^{-6} , inside the matching window; none improves by more than the window. In coordinates, 18 reproduce the parent to within 10^{-6} on every coordinate, the 1.2×10^{-6} excess among them from radii perturbed below that tolerance; one is its mirror image; and 7 relocate filler circles while keeping the same radius multiset and

the parent’s sum to seven decimals. Every one also inherits the parent’s grid order, so k -inheritance is 26 of 26, and P-MU3 is disconfirmed. The template does not reassert itself when a better in-family parent is present; the model returns that parent. The independent evidence is A_anchor , whose anchor-rate of 7 of 28, 25% (13–43%), falls below P-MU1’s 50% threshold, so the template is not self-stable under its own anchor, and there the model does depart: 17 of 28 move above the anchor, 3 of them to the family argmax value, and 4 fall below it. Grid order is kept by 22 of 28, 79% (60–90%); the 6 that change it all move above the anchor and include the 3 that reach the argmax. The asymmetry is the finding: given the argmax the model stays; given its own anchor it leaves. F-MU1 was registered as the negation of the persistence prediction over exactly this pair of conditions, so it carries no content beyond the rule’s own outcome; the arm registered no other falsifier.

The unconditioned rate on the same instrument. Arm L’s generation 0 (§5) supplies the no-parent reference on the same prompt bytes, channel and wrapper: 4 of 9 valid outputs on the anchor at $N = 13$, 44% (19–73%), and 7 of 8 at $N = 31$, 88% (53–98%), against A_anchor ’s 4 of 12 and 2 of 11 at those cells. Arm L did not run at $N = 21$, so no unconditioned row exists there. The table below carries the rows.

P-MU4, held in two cells of three. The one condition where the template does reassert itself is the one the registration predicted it would. Given the off-family diagonal parent, 20 of 34 valid samples abandon the parent entirely and emit a k^* -grid construction, 59% (42–74%). Per cell, with intervals in the arm’s table below, the prediction holds at $N = 21$, 7 of 13, and at $N = 31$, 12 of 14, and not at $N = 13$, where the count is 1 of 7. The attractor captures the trajectory from a worse parent, and at 0 of 26 from a better in-family one.

Parent	Valid	Anchor-rate (95% Wilson)	Registered thresh- old	Score against parent	k - inher.
None (arm L gener- ation 0, same instru- ment)	17	4/9 at $N = 13$ (44%, [19–73%]); 7/8 at $N = 31$ (88%, [53–98%]); no row at $N = 21$	unconditioned ref- erence, no bar	—	—
B_rival (a packing at the family argmax)	26	0/26 (0%, [0–13%])	DISSOLVES if $\leq$ 20%	26/26 (100%, [87–100%]; bar $\geq$ 50%): 25 equal within 10^{-6} , 1 above by 1.2×10^{-6}	26/26
A_anchor (model’s own anchor)	28	7/28 (25%, [13–43%]); per cell 4/12, 1/5 and 2/11 at $N = 13$, 21 and 31	P-MU1 predicted $\geq$ 50%, not met	17/28 above the anchor, 4/28 below	22/28
$C_offfamily$ (diago- nal chain, sum $\approx$ 0.6)	34	20/34 abandon parent for k^* - grid (59%, [42–74%]); per cell 1/7 [3–51%], 7/13 [29–77%] and 12/14 [60–96%] at $N = 13$, 21 and 31	P-MU4: template reasserts, scored per cell	—	—

Bracketed ranges are 95% Wilson intervals on the rate they follow.

Clearance of the family argmax (post hoc, disclosed). No conditioned output clears it at 10^{-9} . Of the eighteen that exceed it at 10^{-6} , seventeen are B_rival outputs returning their parent’s score within 10^{-6} , and that parent’s own seven-decimal coordinates sum to 3.25×10^{-7} above the closed form at $N = 13$ and 2.63×10^{-7} at $N = 31$; the eighteenth is the A_anchor output at $N = 21$ that moved onto the argmax value. Arm MU’s rows were scored against the anchor, so the clearance question was never asked at registration. Recomputing the family argmax from the closed form gives 1.776142375 at $N = 13$, 2.258883476 at $N = 21$ and 2.748528137 at $N = 31$. Arm MU’s 135 invocations give 88 valid outputs at the primary tolerance and 63 at 10^{-9} . The eighteen sit above it by more than the 5×10^{-8} granularity at which their sums are recorded and by at most 1.53×10^{-6} , and each fails the 10^{-9} test on an overlapping pair of circles, not on containment: their largest pairwise overlaps run from 3.8×10^{-9} through a median of 4.4×10^{-8} to 4.3×10^{-7} , and the excess is that overlap. Nine further rows exceed it only within that granularity and the remaining 61 of the 88 do not; at 10^{-9} none of the 63 clears.

5 Five loop generations: arm L

Arm MU samples once; the regime a discovery loop runs in iterates, feeding a scored proposal back as the next parent. Arm L runs the minimal generational loop and reports its four registered predictions whether or not they held.

Design. The design was registered before sampling, with both prompt templates hashed and the stepper and scorer committed alongside. Generation 0 is the bare template, sampled fresh inside this arm, byte-identical in prompt to the unconditioned arms of the companion paper and served through the same channel and wrapper. Generations 1 to 5 use arm MU’s registered mutation template verbatim, instantiated with the parent packing and its score. Two discriminating cells, $N = 13$ and $N = 31$, are crossed with two archive regimes. GREEDY retains only the single best valid proposal. DIVERSE retains up to five distinct-valued ones. That gives four lineages of six rounds at population 5, which is 120 invocations. Two of the 120 were rejected by the runtime before reaching the model, and are counted as invalid rather than dropped. The parent for population slot i is $\text{archive}[i \bmod |\text{archive}|]$, so the loop carries no sampling randomness of its own. All four lineages are evaluable under the registered floor.

This is a minimal analogue and the registration says so: no islands, no crossover, no program representation, no evaluator signal beyond the parent’s own score. No claim about any deployed system follows.

Registered predictions. L1 predicted that the modal valid output at generation 0 equals the registered prediction in every lineage. L2 predicted that dissolution survives iteration, so that the pooled on-prediction rate over generations 1 to 5 falls below the generation-0 rate. L3 predicted that the pooled conditioned on-prediction rate is higher under GREEDY than under DIVERSE, on the reading that loop diversity is scaffold-produced. L4 was registered open with no direction predicted: report best-of-run against the family argmax. Falsifier F-L1 fires if the pooled conditioned rate meets or exceeds the generation-0 rate in both cells.

Table 3: Arm L by lineage. Valid outputs over invocations, and on-prediction counts, at generation 0 and across the five conditioned generations, with each lineage’s best scoring valid output, the generation at which that best first appeared, and its cell’s family argmax. Pooled: 11 of 17 on-prediction at generation 0, 65% (95% Wilson 41–83%), against 11 of 49 conditioned, 22% (13–36%).

Lineage	Gen-0 valid / invocations	Gen-0 on-prediction	Conditioned valid / invocations	Conditioned on-prediction	Best of run	Best first reached (gen)	Family argmax
13-GREEDY	5/5	2	4/25	0	1.7761424	0	1.7761424
13-DIVERSE	4/5	2	16/25	5	1.7761424	5	1.7761424
31-GREEDY	4/5	3	18/25	3	2.6104167	5	2.7485281
31-DIVERSE	4/5	4	11/25	3	2.7485281	2	2.7485281
Pooled	17/20	11	49/100	11			

Validity, and what every rate in this arm conditions on. Generation 0 returns 17 valid outputs from 20 invocations; the five conditioned generations return 49 from 100, and the weakest lineage, 13-GREEDY, returns 4 from 25. The conditioned rounds lose validity as well as accuracy, so every rate here is computed over the survivors of that filter: the on-prediction rate of 11 of 49 is a rate among survivors, and over all invocations it is 11 of 100 (6–19%) against 11 of 20 (34–74%) at generation 0. The registration set no validity prediction for the loop, so the validity loss is reported as a post hoc observation and not as a tested claim.

L2 confirmed, with unequal support. Summing the two lineages at each cell in Table 3, the on-prediction rate falls from 7 of 8 at generation 0, 88% (53–98%), to 6 of 29 conditioned, 21% (10–38%), at $N = 31$, where the intervals are disjoint, and from 4 of 9, 44% (19–73%), to 5 of 20, 25% (11–47%), at $N = 13$, where they overlap and the drop is a point estimate. F-L1 did not fire.

L1 was not met. It fails on one lineage. The 13-DIVERSE seed round in Table 3 splits evenly, and the registered tie rule counts the tie against the prediction. The other three lineages pass. The rule set four per-lineage bars, each over the four or five valid proposals its lineage’s seed round returned, rather than one pooled bar per cell; the miss is a miss of that rule, reported as scored and not rescued.

L3 was not met. The conditioned on-prediction rate is 3 of 22 under GREEDY, 14% (5–33%). It is 8 of 27 under DIVERSE, 30% (16–48%). The difference runs opposite to the registered direction and is not separable from chance: Fisher’s exact test on the 2×2 table gives $p = 0.303$ at an odds ratio of 0.375. The comparison is also confounded: the lineages differ in parent identity as well as archive width, and 13-GREEDY contributes the fewest conditioned valid proposals of any. The outcome is a disconfirmation of the registered prediction, not evidence for its converse.

L4: three of the four lineages reach the family argmax exactly, and none exceeds it. The best-of-run and family argmax columns of Table 3 carry the comparison: both lineages at $N = 13$ and 31-DIVERSE land exactly on their cell’s family argmax, and 31-GREEDY stops short of it. When each best first appeared differs: 13-GREEDY’s best is a generation-0 output, one rival-family packing in its unconditioned seed round that the loop never bettered; 13-DIVERSE reaches the value at generation 5 and 31-DIVERSE at generation 2; 31-GREEDY’s best arrives at generation 5 and never reaches it. No valid conditioned output the table counts exceeds the best value the recipe family can express at 10^{-9} , the clearance tolerance. Iteration moves the proposer off the anchor without moving it past the family. One post-sampling correction is disclosed: the registered configuration stored the family argmax to seven decimals while the clearance test compares at 10^{-9} , so a proposal landing exactly on the rival at $N = 31$ tested as 3.7×10^{-8} above the rounded number. The scorer now recomputes the argmax in closed form; that proposal agrees with it to 1.3×10^{-15} , float noise, and clears nothing. Only the 10^{-9} test is affected.

6 Serving path and reasoning budget: arms P and P-D

Arms MU and L change what the proposer is conditioned on; arms P and P-D change how it is served. Both were registered before sampling, are reported in full in both submissions with no row counted twice (Anonymous authors, 2026), and have per-cell tables in Supplement S5.

Arm P: the registered prompt on a bare, pinned serving path. Arm P sends the registered prompts byte-identical through a serving path that exposes and pins temperature. It runs at $n = 15$ per cell across seven square cells, five held-out cells and three code cells, which is 105, 75 and 45 invocations, 225 in all. Of those, 219 returned content; 6 were refused by the serving path and counted invalid at the cell they were issued for, none of them a square cell, and all six are listed in Supplement S5. The arm was registered with competing predictions on mode identity (P-P1, P-P2), on the code-channel clearance (P-P3, P-P4), and a falsifier on the discriminating-cell rate (F-P1). Each carries a number, and Supplement S1 states all five in the registration’s own words, including the arm’s *dead zone*: the 4 of the 7 square cells that neither P-P1 nor P-P2 adjudicates, so an outcome there confirms neither. Validity collapsed. At the seven square cells, 0 of 105 outputs are valid at either tolerance, and all 105 returned content: well-formed lists of the requested count, no prose, no fences, no reasoning tokens. Every failure but three is an overlap; the three are one non-positive radius, one count mismatch and one malformed row. Their median maximum pairwise overlap at $N = 13$ is 0.10, which is fifty times the value window. At the held-out cells 4 of 75 are valid, all four at $N = 50$ and all four summing to 2.5, a full unit below the family argmax there. None clears. The direct-emission predictions are therefore UNSCOREABLE at every cell under the arm’s own floor of five valid outputs, and F-P1 cannot fire, because its denominator, the valid outputs at the four discriminating square cells, is empty. The registration did not anticipate a collapse, and that is reported rather than read as a zero either way. On the code cells 12 of 45 programs are valid, 4 at $N = 13$, 6 at $N = 21$ and 2 at $N = 31$, only $N = 21$ above the floor; none clears the family, so P-P3 holds.

Two post hoc diagnostics bound the cause. Neither was registered. The first is a temperature sweep: at temperature 0.0, 0.5 and 1.0 the $N = 13$ prompt gives 0 of 6 valid at each, which is 0 of 18 across the three settings. The second is a same-day control at the same cell through the agent runtime, tabled beside D1 and D2 in Supplement S5.

Arm P-D: which component of the instrument. Arm P-D was registered after an outside review asked for a diagnostic beyond temperature, and the registration was pushed before the first call. It sends the same $N = 13$ prompt, to the same alias, on the same pinned path and decoding, under two one-line manipulations, at $n = 15$ each. D1 turns the vendor’s extended thinking on at the routing layer’s default effort and sends no system prompt. D2 leaves thinking off and adds one system line asking for care. The predictions compete.

P-PD1 says the reasoning budget is the cause, and predicts D1 valid in at least 8 of 15 with D2 valid in at most 3. P-PD2 says the missing system context is the cause, with the two thresholds exchanged between D1 and D2. P-PD3 covers anything else. The arm closed at 27 of 30 rows by a registered amendment, after the serving path refused the last three for credit. Those rows could not have changed the verdict, because D2 could reach at most 2 valid while D1 already held 12. D1 is valid in 12 of 14, 86% (95% Wilson 60–96%); D2 in 0 of 13, 0% (0–23%), all overlaps, exactly arm P’s collapse. P-PD1 holds. Validity returns, and the template is the plurality value: $T(4, 13) = 1.625$ at 5 of 12 valid D1 outputs, 42% (19–68%), and grid order $k = 4$ is modal at 6 of 12, so the secondary prediction S-PD1 is met. Supplement S5 accounts for the sixth. Only $N = 13$ was run under D1 and D2, at 14 and 13 rows, so the component is named on one cell and no interval is computed on those counts.

7 Other vendors: arms GM, GM2, GM3 and V

A fourth change of setting is the vendor. Three arms, GM, GM2 and GM3, run against a second vendor’s first-party API with pinned prompts, a fixed output budget and a logged provider per row, and vary the model family; a fourth, arm V, varies vendor and serving class at once. The per-cell tables and the chain’s verdict table are in the supplement.

Arms GM and GM2 returned no scoreable data, and both are reported. The chain registration fixes one set of definitions and predictions for all three arms, over a floor of three valid samples per cell. The primary prediction requires a mode match in at least 5 scoreable cells, the second a pooled on-prediction rate of at least 30%, and the third the two-radius signature at order k^* , a grid radius $1/(2k^*)$ carrying a filler radius $(\sqrt{2} - 1)/(2k^*)$, in at least half of on-prediction samples. The falsifier fires on a mode-match failure in at least 4 scoreable cells. Arm GM addressed a gemini-flash-lite alias at seven cells by twenty samples, 55 rows on the first-party API and 85 through a routed path under an amendment registered before resumption (Supplement S2). It returned 20 valid of 140, under three per cell on average against the floor of three, and was declared UNDERPOWERED under both accountings: 3 scoreable cells with 1 mode match on all rows, 1 cell with none on first-party rows alone, and the falsifier fires under neither. Arm GM2 addressed the gemma family at a 4,096-token output budget. It returned 0 of 140 parseable. The cause is a budget confound, not a model result: the vendor API returns deliberation on a separate channel, and the 4,096 tokens were spent there before any answer text. Arm GM3 re-ran the same design at 16,384 tokens, so GM2 and GM3 differ only by output budget at one vendor, and the same model parses at 123 of 140. Of its 140 rows, 17 did not parse and 80 of the 123 that did are valid packings at 10^{-6} .

Arm GM3: the pooled bar is met and the discriminating cells are not. Arm GM3 runs the unconditioned-call protocol against gemma-4-26b through the vendor’s first-party API, at seven N cells by twenty samples, with predictions and falsifier fixed before sampling. One serving-path deviation is disclosed: the first-party key reached its quota, so 41 of the 140 rows were sampled through a routed path under an amendment committed before resumption, and both accountings are reported. The pooled prediction held. Of the 80 valid packings, 46 land within 2×10^{-3} of the value the closed form predicts, 57.5% (95% Wilson 47–68%), against a registered 30% bar. A pooled rate cannot separate family search from template anchoring, which agree at a non-discriminating cell; only the discriminating cells do, by a rule fitted on another vendor’s models before this family was sampled. The 41 routed rows carry 2 of the 80 valid packings and none of the 46 on-prediction, so first-party rows alone give 46 of 78 (59.0%) against the mixed set’s 46 of 80 (57.5%), and no verdict changes. The pass rides on the non-discriminating cells. 35 of the 46 on-prediction packings sit in the two cells where prediction equals the family argmax. On the three scoreable discriminating cells the rate is 11 of 43, 26% (15–40%), below the bar the pooled metric passes. The cell-level prediction did not hold: the predicted value is modal in 2 of 5 scoreable cells, short of the registered 5, and Figure 1 plots the per-cell modal sums against that prediction. The falsifier did not fire either, because it needed failure in at least 4. The pass bar and the falsifier leave one to three failures unadjudicated, and the observed three fall there: misses of the registered rule, reported as that rule defines. The structural prediction came in under its bar: 20 of 46 on-prediction samples carry the exact two-radius decomposition, against a bar of half. That bar cannot be met at a truncate cell, whose prediction carries one radius and no filler, so the 15 on-prediction samples at $N = 35$ are counted against a signature they could not show.

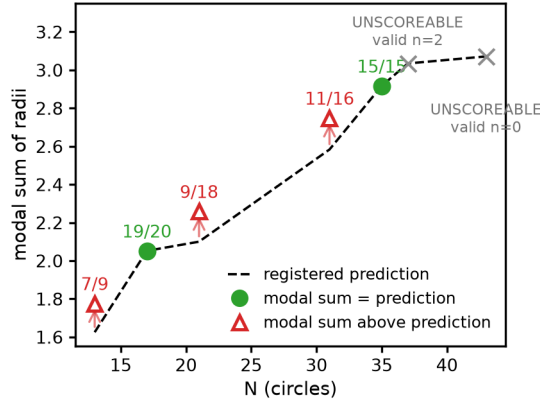


Figure 1: *Arm GM3 modal sums against the registered prediction.* Filled green circles match the predicted mode (the dashed line is $V(k^*, m)$ at extend cells and $T(k^*, N)$ at truncate cells), with frequencies annotated. Open red triangles miss, and every miss lands above the prediction at the family argmax *value*. That these packings are also the family argmax *construction* is established by the post hoc structural check in the text, not by this figure. The two UNSCOREABLE cells are grey crosses on the prediction line, annotated with their valid counts. Regenerates via `fig_gm3_anchoring.py` from `arm_gm3_report.json`.

The misses are the family argmax construction (post hoc, disclosed). The three cell-level misses are not noise. The modal output at every discriminating cell is the family argmax itself. A post hoc structural diagnostic, read-only over the ledger, finds all 27 of the 27 packings at the family argmax value carrying its exact two-radius decomposition, which is the (k^*-1) grid radius with the (k^*-1) filler radius. The registered signature test could not see this, because it tests radii against the predicted order k^* . On the discriminating cells 27 of 43 valid packings, 63% (48–76%), are the family argmax construction. Validity collapses at the two largest cells, $N = 37$ and 43 , which bounds the range of N over which this model returns evaluable packings at all; both cells are UNSCOREABLE under the registered rule and claim nothing.

Arm V: thirteen free-tier aliases. Arm V varies vendor and serving class together, sending the GM chain’s prompt files byte-identical at five cells, $N = 13, 17, 31, 35$ and 37 , five samples per model and cell, to free-tier aliases across seven attributed vendors. The registration was committed before sampling. Three amendments, each committed before the sampling it governs, expanded the original three-alias design to thirteen aliases across two serving paths, and raised the output budget after hidden-reasoning truncation emptied three models’ replies. Execution retried each invocation slot until content arrived, per the registered transport-repair rule, so the ledger holds 877 attempt rows over 325 registered slots, of which 314 carry a row. The per-model floor of §3 is evaluated on the slots, so a slot with no row counts against it. Ten of thirteen aliases finished BELOW-FLOOR, including all three originally registered aliases, and one of them returned all 108 of its attempts without content. Three aliases cleared the floor, and their registered verdicts split. V1 predicted at least 50% of a model’s valid outputs on-prediction, and V2 at most 10% at the family argmax. V2 is scored only at the discriminating cells, $N = 13$ and 31 . In this arm, *transfer* denotes only the registered pooled-bar verdict: an alias clears the per-model floor and meets the V1 bar on its valid outputs pooled over all five cells, a claim about a pooled rate and not about the discriminating cells. All three evaluable aliases come from the amendment-added set, so every transfer verdict in this arm rests on aliases registered by amendment 2 (`arm_v_preregistration_amendment2.md`, d579797).

alias	vendor	valid	V1 on-prediction	V1 bar ($\geq 50\%$)	V2 family argmax valid at discriminating cells (bar: $\leq 10\%$)
north-mini-code	Cohere	12	8/12 [39–86%]	MET (point est.)	0/4 HOLDS
gpt-oss-20b	OpenAI	18	11/18 [39–80%]	MET (point est.)	0/9 HOLDS
gemma-4-31b-it	Google	15	0/15 = 0% [0–20%]	NOT MET	0/3 HOLDS

Bracketed ranges are 95% Wilson intervals on the rate they follow. V2 is scored only at the discriminating cells. The registration labels a met V1 bar TRANSFERS and a missed one DOES-NOT-TRANSFER; this table reports the bar outcome, in the sense §7 defines. Both passing intervals span the 50% bar.

Both passes are weaker than the percentages suggest, in three stated ways. The bracketed Wilson 95% intervals include the bar, so each verdict is a point-estimate pass at $n = 12$ and $n = 18$. Three of thirteen aliases were evaluable at all. And the pooled rate spans cells that cannot discriminate anchoring from family search. Decomposing onto the discriminating cells gives 3 of 4 for the first alias and 5 of 9 for the second. Those denominators are valid outputs at those cells and not registered slots. The arm’s strongest reading,

that these aliases behave as the original family does, does not survive the template-shape null. That null is a proposer choosing uniformly among one branch value per grid order $k \in \{k^*-1, k^*, k^*+1\}$, which lands on-prediction one time in three; Supplement S5 derives it in this paper’s own terms and gives the tails, and this paper applies it post hoc (Anonymous authors, 2026). The pooled rates clear that null, at exact upper tails of 0.019 and 0.014. They clear it on cells where prediction equals the family argmax. At the cells that can discriminate, neither pass separates: 3/4 gives 0.111, 5/9 gives 0.145, and arm GM3’s 11/43 gives 0.895. The V2 prediction holds everywhere evaluable within this arm: no output lands on the family argmax among the 16 valid outputs at discriminating cells.

Anchoring is not confined to the original family, and not universal. Two further vendors’ models land the registered construction values at point estimates above the bar, so at the pooled level anchoring is not confined to the original family, though the discriminating cells do not settle it; nor is it universal. Under the chain’s first-party instrument, gemma-4-26b emits the family argmax construction in 27 of 43 of its valid outputs at discriminating cells. Under arm V’s free-tier instrument, gemma-4-31b lands neither the prediction nor the rival. Its valid packings are weaker de-novo constructions whose per-cell bests run 0.88–1.85 against per-cell predictions of 1.63–3.03, so the best valid packing falls below the prediction at every one of its cells. Model variant and instrument change together across the two measurements, so the split is not attributable from the pair; either way, no single-model measurement would have seen the boundary.

8 Related work

The loop the proposer is characterized for. Language Model Crossover (Meyerson et al., 2024), ELM (Lehman et al., 2022), EvoPrompt (Guo et al., 2024), EoH (Liu et al., 2024) and LLaMEA (van Stein & Bäck, 2025) established the LLM call as an evolutionary operator over prompts and programs. FunSearch (Romera-Paredes et al., 2024) turned the pattern into a discovery claim, and AlphaEvolve (Novikov et al., 2025) and its successors inherit it, OpenEvolve (Sharma, 2025) among them (Lange et al., 2025; Su et al., 2026; Khrulkov et al., 2025; Cemri et al., 2026; Wang et al., 2025). Every one conditions its proposer on a parent and iterates it, two of the conditions this paper varies; simple baselines (Gideoni et al., 2026) and classical solvers (Berthold et al., 2026) recover much of the reported advantage.

Template convergence under iteration. “Mutation Without Variation” (Gurkan et al., 2026) finds iterated LLM program mutation collapsing onto previously seen structural templates: in 87% of chains, over 93% of mutations revisit a seen form. It runs mutation loops in program space with no closed form and no preregistration, the closest published neighbour to arm L; its result is about structural revisiting, arm L’s second prediction about a value the revisited structures do not cross.

Anchoring, and what kind of object anchors here. Anchoring is studied as a general bias over initial information and as numeric primes dragging point estimates (Huang et al., 2026; Lou & Sun, 2024; Malberg et al., 2025). The modality here differs: a construction template has a computable value where a scalar anchor has only a measurable pull. One level up, LLM outputs concentrate where human ones do not (Chen et al., 2026; Jiang et al., 2025), and work on set-level and diversity-aware selection (Li, 2026; Wang et al., 2026; Yang et al., 2026) and the bin-packing critiques (Herrmann & Pallez, 2026; Sim et al., 2025) converge on the same question about the proposer’s contribution.

Two results that cut against this paper’s framing. Evolutionary search matters empirically (Zhang et al., 2024), so one unconditioned call is a poor stand-in for a loop; and strong LLM optimizers act as local refiners (Zhang et al., 2026b), as a parent-conditioned call does.

Instrument effects and reasoning budget. Larger and more instructable models become less reliable (Zhou et al., 2024), a reasoning model can report a higher hallucination rate alongside higher accuracy (OpenAI, 2025), and reasoning collapses past a complexity threshold (Shojaee et al., 2025). Geometric construction is a regime where nominal capability and executed correctness diverge (Kim et al., 2026; Balunović et al., 2025).

Preregistration lineage. Recipes, outcome-blinded predictions and agent protocols are already preregistered (Thomas et al., 2026; Williams et al., 2026; Vaccaro, 2026; Zhang, 2026), and HindsightBench (Jia, 2026) releases frozen preregistrations of directional aggregate hypotheses. This paper’s arms lock a numeric threshold and a falsifier before each setting was changed.

9 Limitations, reproducibility, deviations

Scale. Arm MU takes one conditioning step and arm L five generations at population 5 with one mutation operator, so scale stays open: a larger loop with islands, crossover and more generations could sustain variation this one cannot, and the two contrary results of §8 are met, not settled.

No instruction-free control. Every conditioned call asks for a larger sum and no call without that instruction was run, so returning the *B_rival* parent unchanged, 26 of 26, cannot be split into recognising an optimum and ignoring the instruction. Only its parent-dependence is established: the same instruction moves 17 of 28 off the model’s own anchor.

Vendor and tier scope. The proposer under test is one vendor’s weak-tier alias except where another is named. The cross-vendor arms move the pooled-level statement, not the cell-level mode structure.

Prompt robustness. Every arm conditions on a single hash-locked prompt, which buys reproducibility, not robustness to rewording; the companion paper’s rewording at two discriminating cells found the same modal construction (Anonymous authors, 2026).

Untested alternatives. Typicality bias in preference-tuned models (Zhang et al., 2026a) predicts concentration on the most typical construction without reference to geometry, and nothing here excludes it. Contamination probes (Carlini et al., 2021; Golchin & Surdeanu, 2024), a format-sensitivity sweep (Sclar et al., 2024) and alignment-induced diversity loss (Kirk et al., 2024) remain untested.

Reproducibility. Every arm’s prompts were hashed and its predictions and falsifiers fixed before sampling, with the exceptions the supplement’s deviation table records; no arm was selected on its result. The anonymized artifact (ledgers, scripts, registrations, amendments, frozen reports) regenerates every number by a documented command. Arms P and P-D can be re-sampled on their pinned path; arms MU and L can be re-scored but not re-drawn, because their rows depend on a dispatch wrapper and inherited context the digests do not lock.

Where the forking paths are. Five choices were made with data already in hand, and each is named here. (a) Arm V’s evaluable set exists because of amendment 2 (d579797), which expanded the alias list after the three registered aliases finished below the floor. (b) Arm P-D was designed after arm P collapsed and registered at 5554cc0. (c) The arm MU clearance reading is a post hoc re-read, never registered for that arm. (d) Both arm P diagnostics, the temperature sweep and the same-day runtime control, were chosen after the collapse. (e) Arm P’s scoring code was committed at 596b113, after its registration at aa473ad and after sampling. None is corrected for multiplicity; the protection is that each is listed.

Deviations, in summary. Full rows are in the supplement. Arm GM has one registered serving-path deviation over 85 of 140 rows and arm GM3 one over 41 of 140, both accountings reported for each, with a disclosed duplicate-row repair on GM3; arm V has three pre-sampling amendments; arm L has one post-sampling scorer correction affecting only the 10^{-9} clearance test; and arm P-D closed at 27 of 30 rows by an amendment that could not change its verdict.

Use of AI systems. Claude models drafted prose and assisted implementation; the authors designed the study, specified the instruments and their decision rules, and approved each preregistration before sampling, each commit a git ancestor of the sampling it governs.

10 Conclusion

Eight registered arms changed four things about the setting, for one task and one weak-tier proposer. One conditioning step dissolved the anchor, 0 of 26 outputs on it under a better parent and 7 of 28 under the model’s own, while an off-family parent brought the template back in 20 of 34; five loop generations did not restore it, a pinned serving path collapsed validity until extended thinking was on, and at a second vendor the pooled bar held while the discriminating cells did not. A zero-shot characterization therefore licenses no prediction about a proposer inside a discovery loop without a measurement at that loop’s own conditions. The boundary is two-sided: the anchor does not transfer, the family argmax does, as a ceiling no arm clears.

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